Unit: 6 Unit 4: Absolute Value **Name:** ________________
Topic: Solving Absolute Value Equations **Date:** ____________

Quiz (Carolina Reaper)

Q1. Solve for x: $|x| = 9$
(A) $x = 9$
(B) $x = -9$
(C) $x = 9$ or $x = -9$
(D) $x = 0$
(E) $x = 18$

Q2. Solve for x: $|2x - 5| = 11$
(A) $x = 2$ or $x = 8$
(B) $x = -2$ or $x = -8$
(C) $x = 2$ or $x = -8$
(D) $x = 8$ or $x = -2$
(E) $x = 0$

Q3. A delivery truck travels $|x - 20|$ miles in 2 hours. If the truck's speed is 10 mph, what is x?
(A) $x = 10$
(B) $x = 20$
(C) $x = 30$
(D) $x = 40$
(E) $x = 50$

Q4. Solve for x: $|x + 2| = |3x - 1|$
(A) $x = 1$ or $x = -1$
(B) $x = 1$ or $x = 3$
(C) $x = -1$ or $x = 3$
(D) $x = 0$
(E) $x = 2$

Q5. A shipment of packages must travel $|x + 15|$ miles to reach its destination. If the shipment has already traveled 10 miles, what is x?
(A) $x = -5$
(B) $x = -10$
(C) $x = -15$
(D) $x = -20$
(E) $x = -25$

Q6. Solve for x: $|x| = |2x - 3|$
(A) $x = 1$ or $x = 3$
(B) $x = 1$ or $x = -3$
(C) $x = -1$ or $x = 3$
(D) $x = 0$
(E) $x = 2$

Q7. The time it takes for a package to be delivered is given by $|x - 5| + 2$ hours. If the package is delivered in 4 hours, what is x?
(A) $x = 1$
(B) $x = 3$
(C) $x = 5$
(D) $x = 7$
(E) $x = 9$

Q8. Solve for x: $|x + 1| = |x - 4|$
(A) $x = 1.5$
(B) $x = 2.5$
(C) $x = 3$
(D) $x = 3.5$
(E) $x = 4$

Open Ended Questions (FRQ)

Q9. A traffic flow model is given by the equation $|2x - 10| + 5 = |x + 2| + 11$. Solve for x and show all work.

Q10. A logistics company has a shipping route that is $|x + 20|$ miles long. The company wants to add a new warehouse that is $|x - 5|$ miles from the current route. If the new warehouse is 15 miles from the current route, what is x? Show all work and explain your reasoning.

Chef's Tips

- Emphasize the importance of checking solutions for extraneous results
- Encourage students to use different methods to solve absolute value equations
- Remind students to consider the context of the problem when solving absolute value equations